Calculus Quiz: Assorted Review **Bartonian ID _____**

Write neatly ∘ Boxes are reserved for *final answers only* ∘ *Bartonian ID* must be written on each page ∘ Do not staple.

Complete the following and show work.

(1) Find the one-sided limit of: $\lim_{x \to 1^-} \frac{x^2+2x+1}{x-1}$

1

(2) Find the derivative of the function using the limit method: $f(x)=3x^2+\frac{5}{x}+3$

2

(3) Solve without using a power reduction formula: $\int_0^2 \int_0^{\frac{1}{2}} \cos^2(\theta)\, d\theta\, dr$

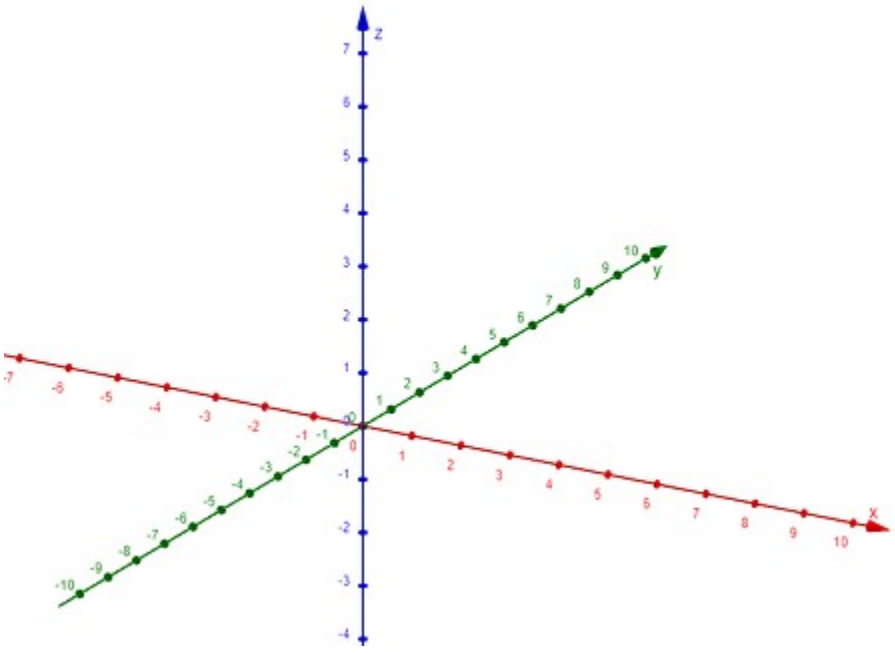
3

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(4) Classify and sketch the quadric surface: $z^2 - x^2 - \frac{y^2}{4} = 1$

4



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5) Prove that this series either converges or diverges? $\sum_{n=1}^{\infty} \frac{2n}{\sqrt{n^2+1}}$

5